A Bayesian Coalescent Model of the DNA Barcode Gap

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Abstract

A simple statistical model of the DNA barcode gap is outlined. Specifically, accuracy of recently introduced nonparametric metrics, inspired by coalescent theory, to characterize the extent of proportional overlap/separation in maximum and minimum pairwise genetic distances within and among species, respectively, is explored in both frequentist and Bayesian contexts. The empirical cumulative distribution function (ECDF) is utilized to estimate probabilities associated with positively skewed extreme tail distance distribution quantiles bounded on the closed unit interval [0, 1] based on a straightforward binomial count of overlapping specimen records. Statistical moments, notably, the mean and variance, are derived for edge cases of no overlap/complete separation and full overlap/no separation in genomic sequence variation, and are shown to be quite tight as sample sizes increase. Further, a new way to visualize distances and the DNA barcode gap estimators via accumulated probabilities appears revealing.

Using R and the probabilistic programming language, Stan, the proposed maximum likelihood estimators and Bayesian model are demonstrated on cytochrome *b* (CYTB) gene sequences from two *Agabus* diving beetle species, *A. bipusulatus* and *A. nevadensis*, exhibiting limits in the extent of representative taxonomic sampling ($N = 701$ and $N = 2$ individuals, respectively). Large-sample theory and MCMC simulations clearly expose problems, showing much uncertainty in parameter estimates, particularly under the former paradigm, and when specimen sample sizes for target species are small. Findings herein highlight the promise of the Bayesian approach using a conjugate beta prior for reliable posterior estimation over classical inference when available data are sparse. Obtained results can help shed light on foundational and applied research questions concerning DNA-based specimen identification and species delineation for studies in evolutionary biology and ecology, as well as biodiversity conservation, forensics, management and restoration, of wide-ranging taxa.

Keywords: Bayesian/frequentist inference, DNA barcoding, Hamiltonian Monte Carlo (HMC), intraspecific genetic distance, interspecific genetic distance, multispecies coalescent, specimen identification, species discovery

1 Introduction

The routine use of DNA sequences to support broad evolutionary hypotheses and questions concerning central demographic processes, like gene flow and speciation, is now ubiquitous. Since the late 1980s, mitochondrial DNA (mtDNA) has been employed to produce a distinctive and measurable signature of genetic polymorphism that readily elucidates phylogeographic patterns, such as colonization and extinction, in diverse and spatially-distributed taxonomic lineages, such as birds, fishes, insects, and arachnids, among other extensively studied groups (Avise et al., 1987). The application of genomic data to applied fields, like biodiversity forensics, conservation, management, and restoration, for the molecular identification of unknown organismal samples, came later (*e.g.*, Forensically Informative Nucleotide Sequencing (FINS); Bartlett and Davidson (1992)). Since its inception over 20 years ago, DNA barcoding (Hebert et al., 2003a) built significantly on earlier work and has emerged as a robust method of specimen identification and species discovery across myriad multicellular eukaryotes. This endeavour has dramatically expedited millions of collected specimens to be sequenced at easily obtained short, standardized gene regions like the cytochrome *c* oxidase subunit I (5'-COI) mitochondrial locus for animals (Hebert et al., 2003b) in the largest taxon-oriented initiative to date. However, despite now far surpassing the depth and breadth of traditional morphospecies taxonomy at 10 times the speed, and for a fraction of the cost, DNA barcoding is not without its controversies.

The success of the single-locus approach, particularly for regulatory applications, depends crucially on two important factors: (1) the availability of high-quality specimen records found in public reference sequence databases such as the Barcode of Life Data Systems (BOLD; http://www.barcodinglife.org) (Ratnasingham and Hebert, 2007) and GenBank (https://www.ncbi.nlm.nih.gov/genbank/), and (2) the establishment of a DNA barcode gap — the notion that the maximum genetic distance observed within species is much smaller than the minimum degree of marker variation found among species (Meyer and Paulay, 2005; Meier et al., 2008). Early research has demonstrated that the presence of a DNA barcode gap hinges strongly on extant levels of species haplotype diversity gauged from comprehensive specimen sampling at wide geographic and ecological scales (Bergsten et al., 2012; Čandek and Kuntner, 2015). Notwithstanding, many taxonomic groups lack adequate separation in their pairwise intraspecific and interspecific genetic distances due to varying rates of evolution in both genes and taxa (Pentinsaari et al., 2016). Furthermore, it has been well-demonstrated that the presence of a DNA barcode gap becomes less certain with increasing spatial scale of sampling, since interspecific distances increase due to population structuring under limited dispersal through isolation-by-distance, while intraspecific distances shrink as more closely-related species are sampled (Phillips et al., 2022). This can pose problems in cases of rare singleton species, endemics, or monotypic taxa, for instance (Ahrens et al., 2016). As a result, rapid matching of unknown query samples to expertly-validated references can be compromised, leading to cases of false positives (taxon oversplitting) and false negatives (excessive lumping of taxa) due to incomplete lineage sorting, interspecies hybridization, genome introgression, species synonymy, cryptic diversity, and specimen misidentifications (Hubert and Hanner, 2015; Phillips et al., 2022). Another culprit includes PCR/sequencing error caused by incorrect nucleotide basecalling, chimeras/heteroplasmies, sample contamination, indels (insertion/deletions), NUMT (nuclear-mitochondrial insert) pseudogene amplification, or *Wolbachia* endosymbiont infection, among other explanations (Phillips et al., 2023). Each of these can artificially magnify levels of genetic diversity in taxon populations, rendering techniques like DNA barcoding unreliable for certain groups like marine invertebrates, which are both widely undersampled and possess slow rates of molecular evolution (Bucklin et al., 2011).

Recent work has argued that DNA barcoding, in its current form, is lacking in statistical rigor (Phillips et al., 2022). Most publications rely strongly on heuristic distance-based thresholds calculated from convenient specimen sample sizes (usually 5-10 specimens per species, but singleton/doubletons are more prevalent). Instead, focus should be placed on employing optimal sampling designs using collated taxon haplotype information, coupled with project funding and budget constraints (Phillips et al., 2015, 2019, 2020). This is necessary to accurately infer taxonomic identity and delineate boundaries on the basis of clustering genetic variation into putative groups reminiscent of species (Phillips et al., 2022). Of these studies, few report measures of uncertainty, such as standard errors (SEs) and confidence intervals (CIs), around estimates of intraspecific and interspecific variation (*e.g.*, Wiemers and Fiedler (2007)), calling into question the existence of a true species' DNA barcode gap (Čandek and Kuntner, 2015; Phillips et al., 2022). To support this idea, novel nonparametric locus- and species-specific metrics based on the multispecies coalescent (MSC) were recently outlined by Phillips et al. (2024) (see **Methods**).

The basic coalescent (Kingman, 1982a,b) encompasses a backwards continuous-time stochastic Markov process of allelic sampling among lineages under the infinite sites model of mutation within a large neutrally-evolving, panmictic, diploid species population of constant size (*i.e.*, obeying Wright-Fisher dynamics) from the present into the past towards the most recent common ancestor (MRCA). Although the coalescent sampling process plays a foundational role in evolutionary biology and population genetics theory, due to its inherent simplicity and flexibility (Rosenberg and Nordborg, 2002; Degnan and Rosenberg, 2009), the approach has been both under-utilized and under-appreciated as a key player within DNA barcoding (Collins and Cruickshank, 2014; Dowton et al., 2014; Hubert and Hanner, 2015; Phillips et al., 2022). Numerous theoretical and empirical investigations have shown that protein-coding sequence variation within non-recombining haploid loci like COI is generally subject to strong purifying selection and selective sweeps at nonsynomonous sites, rather than the occurrence of functionally silent substitutions within synonymous regions; consequently, DNA barcoding works in practice, but not in theory (Stoeckle and Thaler, 2014). However, because mitochondrial markers have considerably smaller effective population sizes (N_e) compared to nuclear regions of the genome, accumulated mutations become fixed, and therefore diagnostic of species, at a much faster rate due to genetic drift (Hubert and Hanner, 2015; Phillips et al., 2022). Previously proposed MSC algorithmic approaches (of which there are too many to exhaustively list here, and which have mostly been applied to multilocus sequence data), generally assume either a strict or relaxed molecular clock and a simplified model of DNA nucleotide substitution across closely-related taxa (Ho and Duchêne, 2014; Flouri et al., 2022). From here, an estimated species phylogeny may be easily constructed (*e.g.*, with or without use of a guide tree) (*e.g.*, Rannala and Yang (2003, 2017); Yang and Rannala (2010, 2014, 2017)). Even tree- and network-based taxon delimitation methods designed for single loci, such as the Generalized Mixed Yule Coalescent (GMYC) (Pons et al., 2006; Monaghan et al., 2009; Fujisawa and Barraclough, 2013), Poisson Tree Processes (PTP) (Zhang et al., 2013; Kapli et al., 2017), and statistical parsimony/haplotype networks (*e.g.*, TCS networks (Templeton et al., 1992), haplowebs (Dellicour and Flot, 2015, 2018)), which see much use in DNA barcoding initiatives, have their flaws. Optimal performance of these methods relies heavily on careful parameter selection (*e.g.*, prior specification of branching rates in ultrametric trees in the case of GMYC, or the probability of parsimony for nested clade analysis (NCA; Templeton (1998)), to name a few) (Hubert et al., 2024) within third-party software like BEAST (Bouckaert et al., 2019), among other concerns (Talavera et al., 2013; Tang et al., 2014; Fonseca et al., 2021).

Hubert et al. (2024) categorized statistical species delimitation methods according to five crucial factors necessary for success: availability, reliability, scalability, understandability, and usability. Any proposed approach should score high across all of these aspects (Hubert et al., 2024). Yet, numerous tradeoffs exist, and so current methods fall short. (See Table 1 in Hubert et al. (2024) for a comprehensive overview.) Careless parameterization of established approaches by users unfamiliar with their underlying statistical characteristics may severely bias overall results in unintended ways (Hart and Sunday, 2007), even if default settings are employed. In contrast, Phillips et al.'s (2024) approach is both tree- and network-free, and does not require judicious parameter setting; all that is needed is a distance matrix. Therefore, the method is easy to use and interpret, quite memory efficient, very fast to run on large datasets, and archived freely online through GitHub (https://github.com/jphill01/DNA-Barcode-Gap-Coalescent). Most importantly, the approach ranks highly across all criteria mentioned by Hubert et al. (2024).

The estimators from Phillips et al. (2024) represent a clear improvement over simple, yet arbitrary, distance heuristics such as the 2% rule noted by Hebert et al. (2003a) and the 10× rule (Hebert et al., 2004). The 2% rule asserts that DNA barcodes differing by at least 2% at sequenced genomic regions should be expected to originate from different biological species. In contrast, the 10× rule suggests that barcode sequences displaying 10 times more genetic variation among species than within taxa is suggestive of a distinct evolutionary origin. These conventions form the core of threshold-based single-locus species delimitation tools like Automatic Barcode Gap Discovery (ABGD) (Puillandre et al., 2011), Assemble Species by Automatic Partitioning (ASAP) (Puillandre et al., 2021), and the Barcode Index Number (BIN) framework (Ratnasingham and Hebert, 2013). However, the lack of adoption of an explicit and universally agreed-upon species concept in DNA barcoding studies is missing. The ideal concept is one that readily governs lineage formation and proliferation necessary to establish rigorous taxon definitions for successful delimitation of hypothesized and heuristic evolutionary units using DNA barcode identification criteria (de Queiroz, 2007; Pante et al., 2015; Rannala, 2015; Wells et al., 2022). Such a concept must be proposed in conjunction with secondary lines of evidence (*e.g.*, morphology, ecology, geography, ontegeny, and behaviour) promised by an integrative framework (Dayrat, 2005). This is the case, for instance, with species existing in allopatry/sympatry, where establishment of reproductive isolation on the basis of fixed diagnostic differences using the Biological Species Concept (BSC) (Mayr, 1942), is challenging (Fujita et al., 2012). Notably, Fujita et al. (2012) remarks that the MSC applied to the taxon delimitation problem harmonizes many species concepts since it identifies independent evolutionary trajectories under the umbrella of the General Lineage Concept. This makes the MSC a fitting tool to model speciation and other modes of diversification (Collins and Cruickshank, 2014). In addition, the reliance on visualization approaches, such as frequency histograms, dotplots, and quadrant plots, to expose DNA barcoding's limitations, has also been criticized since they can fog real species signal relative to noise (Collins and Cruickshank, 2013; Phillips et al., 2022). Up until the work of Phillips et al. (2024), the majority of studies (*e.g.*, Young et al. (2021)) have treated the DNA barcode gap as a binary response. Given poor sampling depth for most taxa, a Yes/No dichotomy is inherently flawed because it can falsely imply a DNA barcode gap is present for a taxon of interest when in fact no such separation in genetic distances exists. On the other hand, in the case of sequence overlap, while the presence a single specimen record is a necessary requirement to nullify the existence of a DNA barcode gap in light of available genetic data, it is not a sufficient condition, since said taxon record could represent a species misidentification or other error. Thus, the MSC serves as an ideal model to settle ongoing debates and issues regarding DNA barcoding's use as both a molecular identification and delineation tool. The recently introduced DNA barcode gap metrics go some way into further operationalizing this task.

Phillips et al.'s (2024) proposed statistics quantify the extent of asymmetric directionality of proportional distance distribution overlap/separation for species within well-sampled taxonomic genera based on a straightforward distance count, in a similar vein to established measures of statistical similarity such as the Kullback-Leibler (KL) divergence (Kullback and Leibler, 1951) and other related statistics. The metrics can be employed in a variety of ways, including to validate performance of marker genes for specimen identification to the species level (using nonparametric bootstrapping, as in Phillips et al. (2024)). The estimators can further be utilized to assess whether computed values are consistent with fundamental population genetic-level parameters like N_e, mutation rate (μ), migration rate (m), and divergence time (τ), as well as derived ones, such as the population mutation rate, $\theta = 4N_e\mu$, and the fixation index, F_{ST}. This is especially true for species under study within a statistical phylogeographic setting (Knowles and Maddison, 2002; Knowles, 2004, 2009; Mather et al., 2019). Early on, species discovery using DNA barcoding was presumed to only work for reciprocally monophyletic groups, and thus concerned itself with terminal branches of generated phylogenies rather than more basal lineages occurring deeper in hypothesized species trees (Fujita et al., 2012; Mutanen et al., 2016). Furthermore, the occurrence of short branches and large N_e within resolved phylogenies increases the likelihood of deep coalescence (Degnan and Rosenberg, 2009), clouding species delimitations, which often fail or are uncertain in broad parameter space (Carstens et al., 2013; Hickerson et al., 2006; Rannala, 2015). As DNA barcoding is a single-locus approach, it is problematic for evolutionarily young taxa. Incomplete lineage sorting within gene genealogies, leading to the retention of ancestral polymorphisms within populations, is a common phenomenon due to the ongoing stochastic dynamic of mutation generating population variation, and genetic drift driving gene variants to fixation (Degnan and Rosenberg, 2009; Rannala, 2015). The most promising way forward in assessing the validity of Phillips et al.'s (2024) method seems to be through the use of software such as BPP (Bayesian Phylogenetics and Phylogeography), which permits efficient full Bayesian simulations under various MSC models (*e.g.*, MSC-I (MSC with introgression; Flouri et al. (2020)) or MSC-M (MSC with migration; Flouri et al. (2023)), among others) using Markov Chain Monte Carlo (MCMC) for tree parameter estimation (via the `A00` option, for instance) (Flouri et al., 2018), or PHRAPL (Phylogeographic Inference using Approximate Likelihoods) (Jackson et al., 2017a,b), which employs tractable phylogenetic likelihood calculations via the genealogical divergence index (gdi). Other hypothesis-driven approaches to species delimitation model selection, like Bayes Factors (BFs) (Leaché et al., 2014) and Approximate Bayesian Computation (ABC) (Camargo et al., 2012), are also possible. Rigorous statistical methodology has been historically integrated into DNA barcoding previously (Matz and Nielsen, 2005; Nielsen and Matz, 2006), though strictly for the purpose of specimen identification. Phillips et al.'s (2024) approach is unique as it also greatly extends usage to delimit species, while still being grounded strongly in evolutionary theory.

While introduction of the new metrics is a step in the right direction, what appears to be missing is a rigorous statistical treatment of the DNA barcode gap. This includes an unbiased way to compute the statistical accuracy of Phillips et al.'s (2024) estimators arising through problems inherent in frequentist maximum likelihood estimation for probability distributions having bounded positive support on the closed unit interval [0, 1]. To this end, here, a Bayesian model of the DNA barcode gap coalescent is introduced to rectify such issues. The model allows accurate estimation of posterior means, posterior standard deviations (SDs), posterior quantiles (*e.g.*, median) and credible intervals (CrIs) for the statistics given datasets of intraspecific and interspecific distances for species of interest.

2 Methods

2.1 DNA Barcode Gap Metrics

The novel nonparametric maximum likelihood estimators (MLEs) of proportional overlap/separation between intraspecific and interspecific distance distributions for a given species (x) to aid assessment of the DNA barcode gap are as follows:

$$p_x = \frac{\#\{d_{ij} \geq a\}}{\#\{d_{ij}\}} \tag{1}$$

$$q_x = \frac{\#\{d_{XY} \leq b\}}{\#\{d_{XY}\}} \tag{2}$$

$$p'_x = \frac{\#\{d_{ij} \geq a'\}}{\#\{d_{ij}\}} \tag{3}$$

$$q'_x = \frac{\#\{d'_{XY} \leq b\}}{\#\{d'_{XY}\}} \tag{4}$$

where d_{ij} are distances within species, d_{XY} are distances among species for an *entire* genus of concern, and d'_{XY} are combined interspecific distances for a target species and its closest neighbouring species. The notation # reflects a count, such that the numerator of Equations (1)-(4) correspondS to the total number of overlapping specimen distances satisfying the given threshold, whereas the denominator corresponds to the total number of distances in the respective dataset. Distances form a continuous distribution and are easily computed from a model of DNA sequence evolution, such as uncorrected or corrected p-distances (Jukes and Cantor, 1969; Kimura, 1980) using, for example, the `dist.dna()` function available in the `ape` R package (Paradis et al., 2004). Quantities a, a', and b correspond to $\min(d_{XY})$, $\min(d'_{XY})$, and $\max(d_{ij})$, the minimum interspecific distance, the minimum combined interspecific distance, and the maximum intraspecific distance, respectively (**Figure 1**). Note, distances are constrained to the interval [0, 1], whereas the metrics are defined only on the interval $[a/a', b]$.

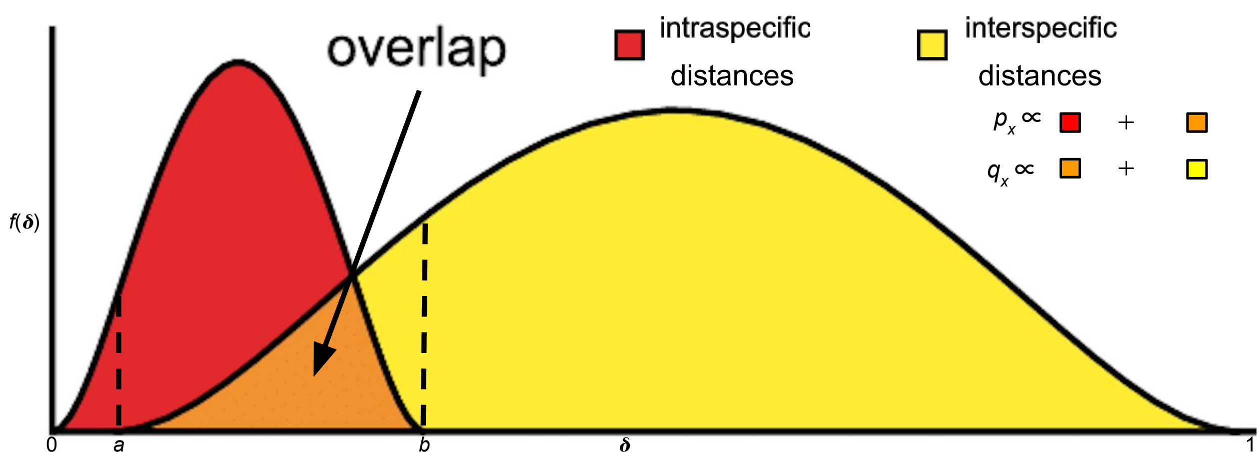

Figure 1: Modified depiction from Meyer and Paulay (2005) and Phillips et al. (2024) of the overlap/separation of intraspecific and interspecific distances (δ) for calculation of the DNA barcode gap metrics (p_x and q_x) for a hypothetical species x. The minimum interspecific distance is denoted by a and the maximum intraspecific distance is indicated by b. The quantity $f(\delta)$ is akin to a kernel density estimate (KDE) of the probability density function of distances. A similar visualization can be displayed for $p_x^{'}$ and $q_x^{'}$ based on intraspecific and combined interspecific distances within the interval $[a', b]$.

Values of the estimators obtained from Equations (1)-(4) close to or equal to zero give evidence for separation between intraspecific and interspecific distance distributions; that is, values suggest the presence of a DNA barcode gap for a target species. Conversely, values near or equal to one give evidence for distribution overlap; that is, values likely indicate the absence of a DNA barcode gap. A simple conceptual diagram to aid interpretation is shown in *Figure 2*.

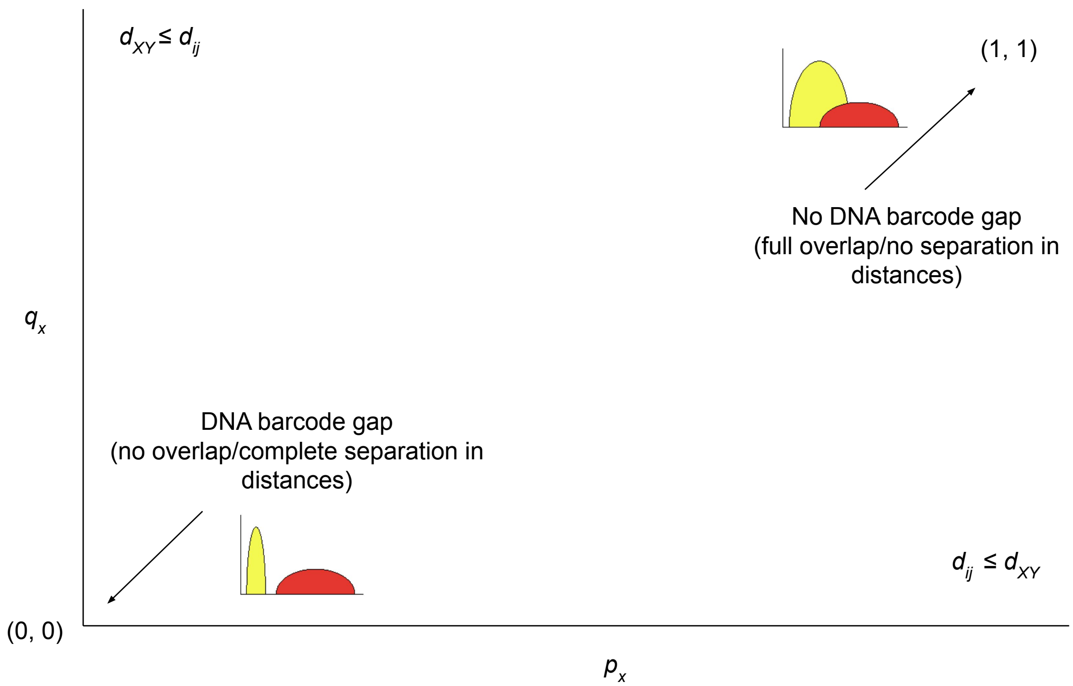

Figure 2: Depiction of the DNA barcode gap metrics and their proper interpretation in cases of no overlap/complete separation and full overlap/no separation in genetic distances.

Furthermore, calculation of the DNA barcode gap estimators is convenient as they implicitly account for total distribution area within $[a/a', b]$ (including the overlap region) (**Figure 1**). Phillips et al.'s (2024) estimators provide a fuller picture of the interaction between intraspecific and interspecific distance distributions because they take advantage of more information when trying to quantify the DNA barcode gap. When attempting to characterize distributions of interest, most scientific studies focus solely on reporting the mean and variance. However, results can be misleading when said distributions possess identical values of these statistical summaries. The metrics of Phillips et al. (2024) offer a deeper perspective in the same way that statistics like skewness (*i.e.*, dstribution asymmetry) and kurtosis (*i.e.*, distribution peakedness/tailedness) It should be noted however that there is a loss of information in computing estimates of Phillips et al.'s (2024) metrics since two different sets of $(p_x,\, q_x)$ and $(p_x^{'},\, q_x^{'})$ values cause equal but distinct underlying distance distributions of $f_{\text{intra}}(\delta)$ and $f_{\text{inter}}(\delta)$ (**Figure 1**). In other words, under certain circumstances, the metrics cannot be determined uniquely, and so are non-identifiable in these cases. An analogy here can be made to Anscombe's quartet, a set of four datasets that differ qualitatively from one another despite having almost identical quantitative statistical summaries. Notice that Equations (1)-(4) are simply empirical partial means of distances falling at and below, or at and exceeding, given distribution thresholds. Notice further that a/a', and b are also the first and nth order statistics, $X_{(1)}$ and $X_{(n)}$, respectively, with $a/a' < b$, which have been pointed out by Phillips et al. (2022) as important for developing a statistical theory to test for the existence of the DNA barcode gap. However, the underlying sampling distributions of $\eta = a - b$ and $\eta' = a' - b$ are not obvious (they involve the evaluation of convolutions). This caveat renders a formal hypothesis test cumbersome to derive since said distributions are not expected to be symmetric. The standard bootstrap resampling procedure (Efron, 1979) can also fail when the estimator of interest is not well-behaved, necessitating extensions like the m-out-of-n bootstrap discussed briefly in Phillips et al. (2022). This is similar to calculating bootstrap support values to assess how well tree nodes justify present data (Efron et al., 1996). However, as more data are collected, support values, along with their associated SEs, will necessarily converge to unity and zero, respectively. Phillips et al. (2024) made an argument for the resampling of alignment sites since DNA barcodes are short in length and sites contributing to distance distribution overlap may not be included within generated bootstrap resamples; yet, no separation could still be evident. Equations (1)-(4) can also be expressed in terms of empirical cumulative distribution functions (ECDFs) (see next section). A clear connection between the ECDF and the nonparametric MLE was noted by Efron and Tibshirani (1993) as well as others. However, calculated values are *not* independent and identically distributed (IID) because estimated standard errors (SEs) will depend on both the number of species sampled within the genus under study, as well as the number of specimens sampled within a target species. To tease this out, Phillips et al. (2024) suggests plotting estimator values against their estimated SEs, along with a simple random downsampling scheme. In the case of two species comprising a focal genus, one well sampled and the other poorly sampled, values of the metrics close to zero for the sufficiently sampled species will likely possess larger SEs following downsizing to match the number of poorly sampled specimens (Phillips et al., 2024). While this scheme appears promising, challenges arise in uncertainty quantification as discussed later.

2.2 Relationship to (Multispecies) Coalescent Theory

The approach of Phillips et al. (2024) differs markedly from the traditional definition of the DNA barcoding gap laid out by Meyer and Paulay (2005) and Meier et al. (2008) in that the proposed metrics incorporate interspecific distances which *include* the target species of interest. Furthermore, if a focal species is found to have multiple nearest neighbours, then the species possessing the smallest average distance is used (Phillips et al., 2024). These schemes more accurately account for species' and coalescent histories inferred from contemporaneous samples of DNA sequences leading to instances of barcode sequence sharing, such as interspecific hybridization/introgression events (Phillips et al., 2024). For example, within Equations (3) and (4), the degree of distance distribution overlap/separation between a target taxon and its nearest neighbouring species, gauged from magnitudes of p'_x and q'_x, is directly proportional to the amount of time in which the two lineages diverged from the MRCA (Phillips et al., 2024). More specifically, $a \propto t_{XY,\,\mathrm{min}}$ and $b \propto t_{ij,\,\mathrm{max}}$, the minimum and maximum (unobserved) divergence time (in units of N_e generations), respectively, for two different individuals (i, j) and species (X, Y) within a well-inventoried genus (Phillips et al., 2024). Figure 1 in Phillips et al. (2024) depicts a phylogeny for two hypothetical species, A and B, showing paraphyly based on the sequencing of two mitochondrial loci. The (unobserved) time the two species formed, t_{AB}, is intermediate between $t_{XY,\,\mathrm{min}}$ and $t_{ij,\,\mathrm{max}}$, but theoretically, its maximum ($t_{AB,\,\mathrm{max}}$) may occur more distantly in the past closer to the root comprising the MRCA for all sampled taxa (Phillips et al., 2024). From a coalescence perspective, p_x is the probability two lineages within species x coalesce after $t_{XY,\,\min}$, whereas q_x represents the probability a lineage within species x coalesces with a lineage in species k (within a sample) after $t_{XY,\,\min}$, but before $t_{ij,\,\max}$. This approach is similar to that of De Sanctis et al. (2022), who devised a coalescent model of taxonomic binning assignment accuracy given query and reference DNA sequences for two species found in a well-curated genomic database. However, the approach herein is inherently simpler, easily extensible, and more intuitive, since it does not make any parametric assumptions regarding mutations along branch lengths, coalescence times, *etc.* and can be applied to more than two taxa in tandem. Thus, the quantities can be used as a criterion to assess the failure of DNA barcoding in recently radiated taxonomic groups, for example, among other plausible biological explanations mentioned earlier.

2.3 The Model

2.3.1 Relating the DNA Barcode Gap Metrics to the Cumulative Distribution Function

Before delving into the derivation of the proposed DNA barcode gap metrics, review of some fundamental statistical theory is necessary. For a given random variable X from a specified continuous probability distribution function (PDF), f, its cumulative distribution function (CDF), F, is defined by

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt = \mathbb{P}(X \leq x) = 1 - \mathbb{P}(X > x). \tag{5}$$

Rearranging Equation (5) gives

$$\mathbb{P}(X > x) = 1 - F_X(x) = 1 - \mathbb{P}(X \leq x), \tag{6}$$

from which it follows that

$$\mathbb{P}(X \geq x) = 1 - F_X(x) + \mathbb{P}(X = x). \tag{7}$$

The CDF has many desirable statistical properties, such as boundedness, continuity, and monotonicity. Consult any graduate-level text on mathematical statistics or probability theory for reference.

Equations (1)-(4) can thus be written in terms of ECDFs via Equations (5) and (7) as follows, since the true underlying Fs, are unknown *a priori*, and therefore must be estimated using available sequence distance data:

$$\begin{aligned} p_x &= \mathbb{P}(d_{ij} \geq a) \\ &= 1 - \hat{F}_{d_{ij}}(a) + \mathbb{P}(d_{ij} = a) \end{aligned} \tag{8}$$

$$\begin{aligned} q_x &= \mathbb{P}(d_{XY} \leq b) \\ &= \hat{F}_{d_{XY}}(b) \end{aligned} \tag{9}$$

$$\begin{aligned} p'_x &= \mathbb{P}(d_{ij} \geq a') \\ &= 1 - \hat{F}_{d_{ij}}(a') + \mathbb{P}(d_{ij} = a') \end{aligned} \tag{10}$$

$$\begin{aligned} q'_x &= \mathbb{P}(d'_{XY} \leq b) \\ &= \hat{F}_{d'_{XY}}(b). \end{aligned} \tag{11}$$

Clearly from **Figure 1**, $\hat{F}_{d_{XY}}(a) = \hat{F}_{d_{ij}}(0) = 0$ and $\hat{F}_{d_{ij}}(b) = \hat{F}_{d_{XY}}(1) = 1$, and, in principle,

$\hat{F}_{d_{XY}}(0) = 0$ and $\hat{F}_{d_{ij}}(1) = 1$. Given n increasing-ordered data points, the (discrete) ECDF, $\hat{F}_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[x_i \leq t]}$, comprises an increasing step function having jump discontinuities of size $\frac{1}{n}$ at each sample observation (x_i), excluding ties (or steps of weight $\frac{i}{n}$ with duplicate observations), where $\mathbb{1}(x)$ is the indicator function. Note that $1 - \hat{F}_n(t)$ is the complementary ECDF (CECDF), which is a declining function. Further, $\mathbb{P}(X = t)$ is the probability that X (a distance) takes on the value t. The ECDF/CECDF has good asymptotic behaviour in the limit of sample size: both functions are bounded on the interval [0, 1]. Equations (8)-(11) clearly demonstrate the asymmetric directionality of the proposed metrics. Plots of the ECDFs/CECDFs for nonzero intraspecific, interspecific, and combined interspecific distances, superimposed with corresponding values of a/a' and b, provide a new, useful way to visualize the DNA barcode gap and Phillips et al.'s (2024) statistics. Firstly, there is no need to tune the output (*e.g.*, number of bins and bin width in the case of histograms (Phillips et al., 2022)). Secondly, probabilities reflecting areas under the distance curves depicted in **Figure 1** can be directly read off the ECDF/CECDF curves. Values of the metrics can also be determined from direct inspection of the ECDF/CECDF plot by determining the corresponding y-value (cumulative probability) for a given value of x (distance).

2.3.2 Integrating Frequentist and Bayesian Inference

A major criticism of large sample (frequentist) theory is that it relies on asymptotic properties of the MLE (whose population parameter is assumed to be a fixed but unknown quantity), such as estimator normality and consistency as the sample size approaches infinity. This problem is especially pronounced in the case of binomial proportions (Newcombe, 1998). The estimated Wald SE of the sample proportion, is given by $\widehat{SE[\hat{p}]} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where $\hat{p} = \frac{Y}{n}$ is the MLE, Y is the total number of successes ($Y = \sum_{i=1}^{n} y_i$), and n is the total number of trials (*i.e.*, sample size). However, the above formula for the standard error is problematic for several reasons. First, it is a Normal approximation which makes use of the central limit theorem (CLT); thus, large sample sizes are required for reliable estimation. When few observations are available, SEs will be large and inaccurate, leading to low statistical power to detect a true DNA barcode gap when one actually exists. Further, resulting interval estimates could span values less than zero or greater than one, or have zero width, which is practically meaningless. Second, when proportions are exactly equal to zero or one, resulting SEs will be exactly zero, rendering $\widehat{SE}[\hat{p}]$ given above completely useless. In the context of the proposed DNA barcode gap metrics, values obtained at the boundaries of their support are often encountered. Therefore, reliable calculation of SEs is not feasible. Given the importance of sufficient sampling of species genetic diversity for DNA barcoding initiatives, a different statistical estimation approach is necessary.

Bayesian inference offers a natural path forward in this regard since it allows for straightforward specification of prior beliefs concerning unknown model parameters and permits the seamless propagation of uncertainty, when data are lacking and sample sizes are small, through integration with the likelihood function associated with true generating processes. The posterior distribution ($\pi(\theta|Y)$) is given by Bayes' theorem up to a constant of proportionality, $\pi(\theta|Y) \propto \pi(Y|\theta)\pi(\theta)$, where θ are unobserved parameters, Y are known data, $\pi(Y|\theta)$ is the likelihood, and $\pi(\theta)$ is the prior. As a consequence, because parameters are treated as random variables, Bayesian models are much more flexible and generally more easily interpretable compared to frequentist approaches. Under the Bayesian paradigm, entire posterior distributions, along with their statistical summaries (*e.g.*, CrIs) are outputted, rather than just long run behaviour reflected in sampling distributions, p-values, and CIs as in the frequentist case, thus allowing direct probability statements to be made.

2.3.3 Relating the DNA Barcode Gap Metrics to Mixture Models

Essentially, from a statistical perspective, the goal herein is to nonparametrically estimate probabilities corresponding to extreme tail quantiles for positive highly skewed distributions on the unit interval (or any closed subinterval thereof). Here, it is sought to numerically approximate the extent of proportional overlap/separation of intraspecific and interspecific distance distributions within the subinterval $[a/a', b]$. This is a challenging computational problem within the current study as detailed in subsequent sections. The usual approach employs kernel density estimation (KDE), along with numerical or Monte Carlo integration of complicated PDFs, and invocation of extreme value theory (EVT) to calculate overlap (such as that for species with unusually large distances suggestive of cryptic species diversity, or deep sequence divergence due to long periods of genetic isolation); however, this requires careful selection of the bandwidth parameter, which would need to be identical for both distributions, among other considerations. This becomes problematic when fitting finite mixture models using MCMC, where nonidentifiability is rampant due to the occurrence of label switching, making clusters hard to discern across sampling iterations, and model selection challenging due to lack of exchangeability among observations (Stephens, 2000). For DNA barcode gap estimation, this would correspond to a two-component mixture (one for intraspecific distance comparisons, and the other for interspecific comparisons), with one or more distribution modes or curve intersection points between components, and the presence of zero distance inflation. This makes parameter estimation difficult using methods like the Expectation-Maximization (EM) algorithm (Dempster et al., 1977) as the algorithm may become stuck in suboptimal regions of the parameter search space and prematurely converge to local optima.

The abovementioned statistical issues seem wholly plausible, especially given that the genus-level distribution of genetic distances is a mixture of closely-related (*i.e.*, nearest neighbour) and more distant species. Examining **Figure 1**, depending on divergence times and other elements, interspecific distributions would be expected to comprise greater density in the centre and upper (right) tail (which make up more distantly-related taxa relative to a target species, and where there are more combinations of distance comparisons possible) compared to the lower tail overlapping with the upper (right) tails of intraspecific distributions (which contain only the closest conspecifics). A key problem affecting reliable inference of the DNA barcode gap metrics is undersampling of genus-level and/or interspecific variation. An immediate, but non-trivial solution appears to be a move toward next-generation DNA barcoding incorporating multiple loci (Collins and Cruickshank, 2014; Dowton et al., 2014). In theory, this would facilitate the detection of missing operational taxonomic units (OTUs), both extant and extinct, when attempting to better characterize a genus. In this context, Phillips et al.'s (2024) statistics could be informative for studies of human phylogeography and ancestry composition where historical admixture between sympatric populations was commonplace (*e.g.*, modern human/Denisova interbreeding). Here, for simplicity, an alternate route is taken to avoid these obstacles.

2.3.4 Bayesian Model Setup

Counts, y, of overlapping distances (as expressed in the numerator of Equations (1)-(4)) are treated as binomially distributed. Given total counts, Y, $\pi(Y|\theta) = \binom{n}{Y}\theta^{y}(1-\theta)^{n-Y}$, where $\binom{n}{Y}$ is the binomial coefficient expressing the number of distinct ways to choose Y elements from n total objects. Y thus has expectation $\mathbb{E}[Y|\theta] = n\theta$, and variance $\mathbb{V}[Y|\theta] = n\theta(1-\theta)$, where $n = \{N, C\}$ are total counts of intraspecific and combined interspecific distances, respectively, for a target species along with its nearest neighbour species, and $n = M$ is a total count for all interspecific species comparisons. This follows from the fact that the ECDF/CECDF is binomially distributed. The quantity thus being estimated is the parameter vector $\underline{\theta} = \{p_x, q_x, p_x^{'}, q_x^{'}\}$. The metrics encompassing $\underline{\theta}$ are presumed to follow a Beta(α, β) distribution, with real shape parameters α and β, which is a natural choice of prior on probabilities. That is, $\pi(\theta|\alpha,\beta) = \frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$, where $\mathrm{B}(\alpha,\beta)$ is the beta function evaluated at α and β. The beta distribution has a prior mean of $\mathbb{E}[\theta|\alpha,\beta] = \frac{\alpha}{\alpha+\beta}$ and a prior variance equal to $\mathbb{V}[\theta|\alpha,\beta] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. In the case where $\alpha = \beta$, all generated Beta(α, β) distributions will possess the same prior expectation, whereas the prior variance will shrink as both α and β increase. Such a scheme is quite convenient since the beta distribution is conjugate to the binomial distribution. Thus, the posterior distribution is also beta distributed, specifically, $\pi(\theta|Y,\alpha,\beta) = \theta^{\alpha+Y-1}(1-\theta)^{\beta+n-Y-1}$

(*i.e.*, Beta$(\alpha + Y,\ \beta + n - Y)$), having expectation $\mathbb{E}[\theta|Y, \alpha, \beta] = \frac{\alpha+Y}{\alpha+\beta+n}$ and variance $\mathbb{V}[\theta|Y, \alpha, \beta] = \frac{(\alpha+Y)(\beta+n-Y)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$. In the context of DNA barcoding, it is important that the DNA barcode gap metrics effectively differentiate between extremes of no overlap/complete separation and full overlap/no separation, corresponding to values of the metrics equal to 0 and 1 (equivalent to total distance counts of 0 and n), respectively. These extremes yield a posterior expectation of $\mathbb{E}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha}{\alpha+\beta+n}$ and a posterior variance of $\mathbb{V}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha(\beta+n)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$, and $\mathbb{E}[\theta|Y = n, \alpha, \beta] = \frac{\alpha+n}{\alpha+\beta+n}$ and $\mathbb{V}[\theta|Y = n, \alpha, \beta] = \frac{(\alpha+n)\beta}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$. Note, the posterior variances are equivalent at these thresholds for all $\alpha = \beta$.

Parameters were given an uninformative Beta(1, 1) prior, which is identical to a standard uniform (Uniform(0, 1)) prior since it places equal probability on all parameter values within its support. This distribution has an expected value of $\mu = \frac{1}{2} = 0.500$ and a variance of $\sigma^2 = \frac{1}{12} \approx 0.083$. Further, the posterior is Beta$(Y+1, n-Y+1)$, from which various moments such as the expected value $\mathbb{E}[\theta|Y] = \frac{Y+1}{n+2}$ and variance $\mathbb{V}|\theta[Y] = \frac{(Y+1)(n-Y+1)}{(n+2)^2(n+3)}$, and other quantities, can be easily calculated. Clearly, $\mathbb{E}[\theta|Y = 0] = \frac{1}{n+2}$ and $\mathbb{V}[\theta|Y = 0] = \frac{n+1}{(n+2)^2(n+3)}$, and $\mathbb{E}[\theta|Y = n] = \frac{n+1}{n+2}$ and $\mathbb{V}[\theta|Y = n] = \frac{n+1}{(n+2)^2(n+3)}$. For example, with a single distance ($n = 1$) for a given species (wherein two specimens are compared), and where the distance is not overlapping, on average, θ will be equal to $\mu = \frac{1}{3} = 0.333$, with a variance of $\sigma^2 = \frac{2}{36} = \frac{1}{18} \approx 0.056$. In contrast, with $n = 100$ individuals and there is no overlap, the mean reduces to $\mu = \frac{1}{102} \approx 0.010$, with a variance of $\sigma^2 = \frac{101}{1071612} = 9.425 \times 10^{-5}$. When $n = 1$ and there is overlap, the mean is $\mu = \frac{2}{3} = 0.667$, having a variance of $\sigma^2 \approx 0.056$. For $n = 100$ where all distances overlap, θ has expectation $\mu = \frac{101}{102} \approx 0.990$ and variance $\sigma^2 \approx 9.425 \times 10^{-5}$. These edge case examples have good asymptotic behaviour with increasing sample size (variance tends to zero), and demonstrate that the Beta(1, 1) distribution serves as a reasonable first-pass prior for DNA barcode gap estimation using Phillips et al.'s (2024) metrics, though tighter bounds can be achieved through increasing both α and β. In general however, when possible, it is always advisable to incorporate prior information, even if only weak, rather than simply imposing complete ignorance in the form of a flat prior distribution. In the case of unimodal distributions, the (estimated) posterior mean often possesses the property that it readily decomposes into a convex linear combination, in the form of a weighted sum, of the (estimated) prior mean and the MLE. That is $\hat{\mu}_{\text{posterior}} = w\hat{\mu}_{\text{prior}} + (1 - w)\hat{\mu}_{\text{MLE}}$, where, for the beta distribution, $w = \frac{\alpha+\beta}{\alpha+\beta+n}$. Therefore, with sufficient data, $w \to 0$ as $n \to \infty$, regardless of the values of α and β, and the choice of prior distribution becomes less important since the posterior will be dominated by the likelihood. For the Beta(1, 1), $w = \frac{2}{2+n}$, with $n = 1$ giving $w = \frac{2}{3} = 0.667$. For $n = 100$, the prior weight becomes $w = \frac{2}{102} \approx 0.020$. The full Bayesian model for species x is thus given by

$$\begin{aligned}
y_{\text{lwr}} &\sim \text{Binomial}(N, p_{\text{lwr}}) \\
y_{\text{upr}} &\sim \text{Binomial}(M, p_{\text{upr}}) \\
y'_{\text{lwr}} &\sim \text{Binomial}(N, p'_{\text{lwr}}) \\
y'_{\text{upr}} &\sim \text{Binomial}(C, p'_{\text{upr}}) \\
p_{\text{lwr}}, p_{\text{upr}}, p'_{\text{lwr}}, p'_{\text{upr}} &\sim \text{Beta}(1, 1)
\end{aligned} \tag{12}$$

Note that p_x, q_x, p'_x, and q'_x in Equations (1)-(4) are denoted p_{lwr}, p_{upr}, p'_{lwr}, and q'_{upr}, respectively within Equation (12) for easy distinction between MLEs and Bayesian posterior estimates. Further, y_{lwr}, y_{upr}, y'_{lwr}, and y'_{upr} are calculated from the numerators of Equations (1)-(4), respectively. The above statistical theory and derivations lay a good foundation for the remainder of this paper.

2.3.5 Computational Details

The proposed model is inherently vectorized to allow processing of multiple species datasets simultaneously. Model fitting was achieved using the Stan probabilistic programming language (Carpenter et al., 2017) framework for Hamiltonian Monte Carlo (HMC) (Neal, 2011) via the No-U-Turn Sampler (NUTS) sampling algorithm (Hoffman and Gelman, 2014) through the `rstan` R package (version 2.32.6) (Stan Development Team, 2023) in R (version 4.4.1) (R Core Team, 2024). Four Markov chains were run for 2000 iterations each in parallel across four cores with random parameter initializations. Within each chain, a total of 1000 samples was discarded as warmup (*i.e.*, burnin) to reduce dependence on starting conditions and to ensure posterior samples are reflective of the equilibrium distribution. Further, 1000 post-warmup draws were utilized per chain during the sampling phase. Because HMC/NUTS results in dependent samples that are minimally autocorrelated, chain thinning is not required. Each of these tuning parameters reflect default MCMC settings in Stan to control both bias and variance, respectively, in the resulting draws. All analyses in the present work were carried out on a 2023 Apple MacBook Pro with M2 chip and 16 GB RAM running macOS Ventura 13.2. A random seed was set to ensure reproducibility of model results. Outputted estimates were rounded to three decimal places of precision. Posterior draws and posterior distributions of the metrics were visualized as scatterplots and KDE plots, respectively, using the `ggplot()` function within the `ggplot2` R package (version 3.5.1) (Wickham, 2016) with the default Gaussian kernel and optimal smoothness selection. To successfully run the Stan program, end users must have installed an appropriate compiler (such as GCC or Clang) which is compatible with their operating system, such as macOS.

Convergence was assessed both visually and quantitatively as follows: (1) through examining parameter traceplots, which depict the trajectory of accepted MCMC draws as a function of the number of iterations, (2) through monitoring the Gelman-Rubin potential scale reduction factor statistic ($\hat{R}$) (Gelman and Rubin, 1992; Vehtari et al., 2021), which measures the concordance of within-chain *versus* between-chain variance, and (3) through calculating the effective sample size (ESS) for each parameter, which quantifies the number of independent samples generated Markov chains are equivalent to. Mixing of chains was deemed sufficient when traceplots looked like "fuzzy caterpillars", $\hat{R} < 1.01$, and effective sample sizes were reasonably large (Gelman et al., 2020). After sampling, a number of summary quantities were reported, including posterior means, posterior SDs, and posterior quantiles from which 95% CrIs could be computed to make probabilistic inferences concerning true population parameters. To validate the overall correctness of the proposed statistical model given by Equation (12), as a means of comparison, posterior predictive checks (PPCs) were also employed to generate binomial random variates in the form of mean counts from the posterior predictive distribution; that is $\underline{\gamma} = \{Np_x, Mq_x, Np_x^{'}, Cq_x^{'}\}$, to verify that the model adequately captures relevant features of the observed data.

2.3.6 Probabilistic Interpretation of the DNA Barcode Gap Metrics

The proposed Bayesian model outlined here has a straightforward interpretation (**Table 1**).

Table 1: Interpretation of the DNA barcode gap estimators within $[a/a', b]$

Parameter	**Explanation**
p_x/p_{lwr}	When p_x/p_{lwr} is close to 0 (1), it suggests that the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average, while the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average; that is, there is (no) evidence for a DNA barcode gap.
q_x/p_{upr}	When q_x/p_{upr} is close to 0 (1), it suggests that the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average, while the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average; that is, there is (no) evidence for a DNA barcode gap.
$p_x^{'}/p_{\text{lwr}}^{'}$	When $p_x^{'}/p_{\text{lwr}}^{'}$ is close to 0 (1), it suggests that the probability of intraspecific (combined interspecific distances for a target species and its nearest neighbour species) distances being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average, while the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average; that is, there is (no) evidence for a DNA barcode gap.
$q_x^{'}/p_{\text{upr}}^{'}$	When $q_x^{'}/p_{\text{upr}}^{'}$ is close to 0 (1), it suggests that the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average, while the probability of intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average; that is, there is (no) evidence for a DNA barcode gap.

2.4 *Agabus* Case Study

The DNA barcode gap statistics have been shown to hold strong promise for reliable DNA barcode gap assessment when applied to the predatory diving beetles *Agabus* (Coleoptera: Dytiscidae) (Phillips et al., 2024). With a Holarctic range, despite their ease of sampling and well-established taxonomy, this group possesses few morphologically-distinct taxonomic characters that readily facilitate their assignment to the species level (Bergsten et al., 2012). Accurate taxon discrimination within *Agabus* is paramount since several species are currently classified as endangered on the International Union for Conservation of Nature (IUCN) Red List (https://www.iucnredlist.org) due to habitat fragmentation/loss, water pollution, and climate change. Further, the proposed metrics indicate that sister species pairs from this taxon are often difficult to distinguish on the basis of their DNA barcode sequences (Phillips et al., 2024), which may be caused by *Agabus*'s increased degree of non-monophyly at wider geospatial scales, occurrence of hybridization/introgression, or presence of synonymy (Bergsten et al., 2012). Using sequence data from three mitochondrial cytochrome markers (5'-COI, 3'-COI, and cytochrome *b* (CYTB)) obtained from BOLD and GenBank, results from Phillips et al. (2024) highlight that DNA barcoding has been a one-sided argument. Phillips et al.'s (2024) findings point to the need to balance both the sufficient collection of specimens, as well as the extensive sampling of species. Not surprisingly, DNA barcode libraries for most taxonomic groups are biased toward the latter effort (Phillips et al., 2015, 2019, 2022).

The *Agabus* CYTB dataset analyzed by Phillips et al. (2024) is revisited herein for the species *A. bipustulatus* and *A. nevadensis*, since these taxa were the sole representatives for this locus, with the most and the least specimen records, respectively ($N = 701$ and $N = 2$) across all three assessed molecular markers. Briefly, using the R package `MACER` (Young et al., 2021), DNA sequences were downloaded from GenBank and BOLD and processed using a stringent workflow to obtain a 343-bp high-quality FASTA alignment representing 46 unique haplotypes. The resulting alignment contained no stop codons, ambiguous nucleotides, or other problematic artifacts that bias genetic diversity estimates. Further, both species- and genus-level distance outliers were excluded based on the interquartile range (IQR) (Young et al., 2021). Genetic distances were calculated using uncorrected p-distances. *A. bipustulatus* comprised 45 total haplotypes, whereas

A. nevadensis possessed only one haplotype. Note, DNA barcode gap estimation is only possible for species having at least two specimen records because only one distance can be calculated. This dataset is a prime illustrative example highlighting the issue of inadequate taxon sampling, which arises frequently in large-scale phylogenetic and phylogeographic studies, in several respects. First, from a statistical viewpoint, sample sizes reflect extremes in reliable parameter estimation. Second, from a DNA barcoding perspective, *Agabus* currently comprises about 200 extant Linnean species according to the Global Biodiversity Information Facility (GBIF) (https://www.gbif.org); yet, due to the level of convenience sampling inherent in taxonomic collection efforts for this genus, adequate representation of species and genetic diversity is far from complete. Indeed, herein, species coverage for CYTB is only around 1.000%. Not surprisingly, CYTB performed poorly as a DNA barcode locus for *Agabus* despite having the highest specimen representativeness for a given species ($N = 701$, *A. bipustulatus*) and displaying good separation of genetic sequences among all three mitochondrial genes examined in Phillips et al.'s (2024) study; yet no barcode gap was observed since all metrics were found to be equal to 1.000.

3 Results and Discussion

ECDF/CECDF plots were generated using the base R function `ecdf()`, along with `ggplot()`, and are displayed in **Figure 3**. Interpretation is simple: in the case of *A. bipustulatus*, for both p_x and $p_x^{'}$, all distances are greater than $a = 0$, with 36.900% of distances equal to a/a'. Conversely, both q_x and $q_x^{'}$ equal one because all interspecific distances are less than $b \approx 0.026$; for *A. nevadensis*, both q_x and $q_x^{'}$ are approximately equal to 0.010 since only about 1% of distances fall below $b = 0$. In the case of p_x and $p_x^{'}$, both statistics equal 1.000 since all distances were found be zero.

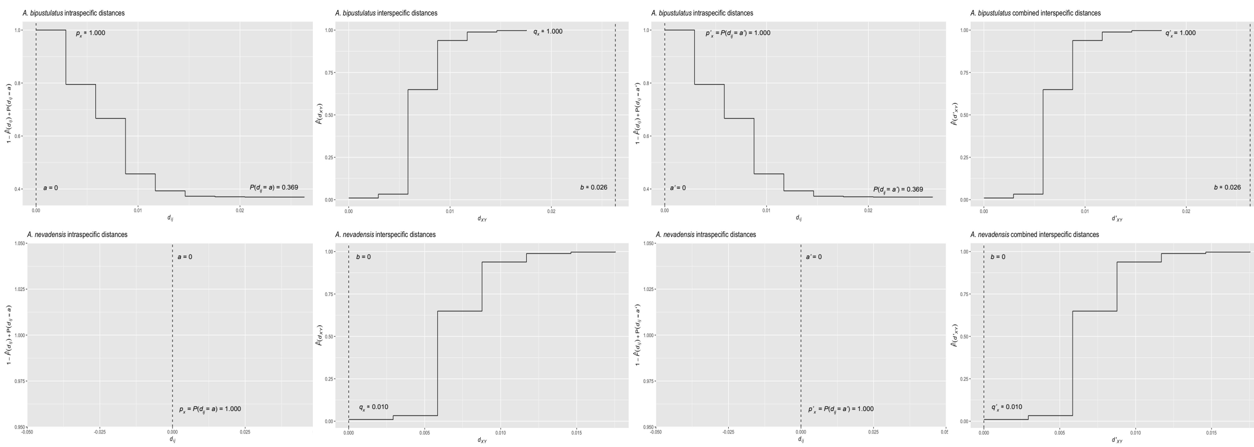

Figure 3: ECDF/CECDF plots of the DNA barcode gap metrics for *A. bipustulatus* and *A. nevadensis* with a/a' and b clearly indicated as dashed vertical lines. When all genetic distances are equal to zero, no step function is shown.

MCMC parameter traceplots showed rapid mixing of chains to the stationary distribution (**Figure 4**). Further, all $\hat{R}$ and ESS values (both not shown) were close to their recommended cutoffs of one and thousands of samples, respectively, indicating chains are both well-mixed and have converged to the posterior distribution. Bayesian posterior estimates were reported alongside frequentist MLEs, in addition to SEs, posterior SDs, 95% CIs, and 95% CrIs (**Table 2**).

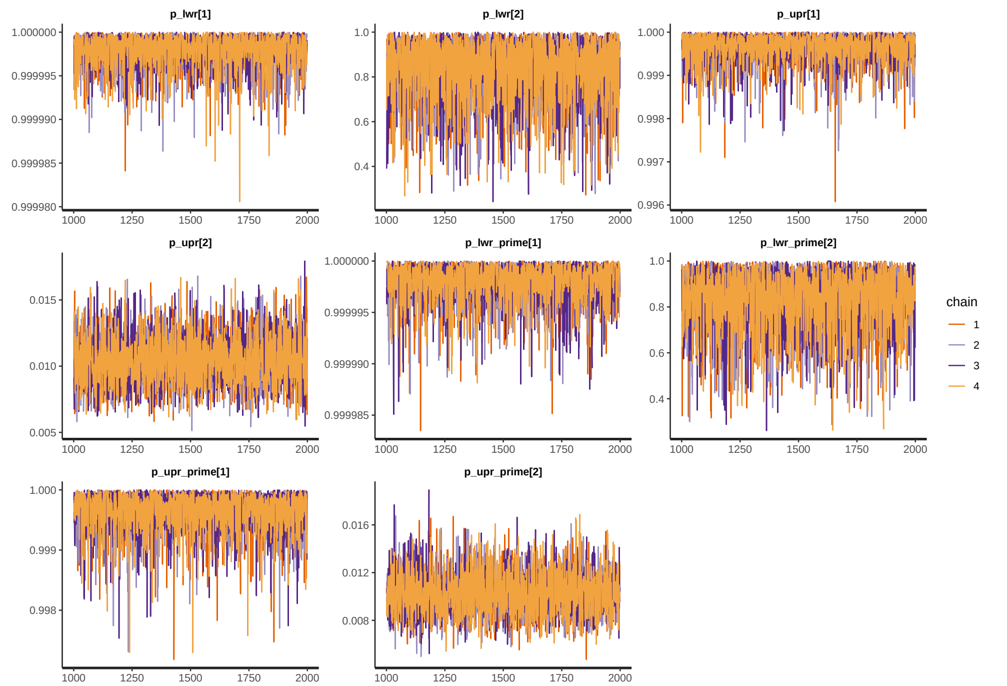

Figure 4: MCMC parameter traceplots applied to *A. bipustulatus* ([1]; $N = 701$) and *A. nevadensis* ([2]; $N = 2$) for CYTB across 1000 post-warmup iterations.

Table 2: Nonparametric frequentist and Bayesian estimates of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) for CYTB, including 95% CIs and CrIs. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

Species	Parameter	MLE (SE, 95% CI)	Bayes Est. (SD; 95% CrI)
A. bipustulatus	p_x/p_{lwr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 1.000-1.000)
A. bipustulatus	q_x/p_{upr}	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 0.999-1.000)
A. bipustulatus	$p_x^{'}/p_{\text{lwr}}^{'}$	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 1.000-1.000)
A. bipustulatus	$q_x^{'}/p_{\text{upr}}^{'}$	1.000 (0.000; 1.000-1.000)	1.000 (0.000; 0.999-1.000)
A. nevadensis	p_x/p_{lwr}	1.000 (0.000; 1.000-1.000)	0.835 (0.144; 0.470-0.996)
A. nevadensis	q_x/p_{upr}	0.010 (0.002; 0.006-0.014)	0.010 (0.002; 0.007-0.014)
A. nevadensis	$p_x^{'}/p_{\text{lwr}}^{'}$	1.000 (0.000; 1.000-1.000)	0.834 (0.138; 0.481-0.994)
A. nevadensis	$q_x^{'}/p_{\text{upr}}^{'}$	0.010 (0.070; -0.128-0.148)	0.010 (0.002; 0.007-0.014)

CIs were calculated using the usual large sample $(1 - \alpha)100\%$-level interval estimate given by $\hat{p} \pm z_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where $z_{1-\frac{\alpha}{2}} = 1.960$ is the critical value for 95% confidence (*i.e.*, the 97.5th percent quantile from the standard Normal distribution), and α is the stated significance level (here, 5%). Given a $(1 - \alpha)100\%$ CI, with repeated sampling, on average

$(1-\alpha)100\%$ of constructed intervals will contain the true parameter of interest, assuming a correctly-specified statistical model; on the other hand, any given CI will either capture or exclude the true parameter with 100% certainty. This in stark contrast to a CrI, where the true parameter is contained within said interval with $(1-\alpha)100\%$ probability, contingent on the data generating model being correct. Note, by default Stan computes equal-tailed (central) CrIs such that there is equal area situated in the left and right tail of the posterior distribution. For a 95% CrI, this corresponds to the 2.5th and 97.5th percent quantiles. However, constructed intervals are usually only valid for symmetric or nearly symmetric distributions. Given the bounded nature of the DNA barcode gap metrics, whose posterior distributions, as expected, show considerable skewness, an alternative approach to reporting CrIs, such as Highest Posterior Density (HPD) intervals (Chen and Shao, 1999) or shortest probability intervals (SPIn) (Liu et al., 2015) is warranted. Although the data for *Agabus* were not found to be consistent with a value of zero for the metrics (suggesting the existence of a DNA barcode gap) for either species, this is clearly a case that must differentiated from one where some or all of the estimators attain a value of one (indicating no DNA barcode gap likely exists). As such asymmetric intervals generally attain greater statistical efficiency (in the form of smaller mean squared error (MSE) or variance) and higher coverage probabilities than more standard interval estimates, careful in-depth comparison is left for future work.

Findings based on nonparametric MLEs and Bayesian posterior means were quite comparable with one another and show evidence of complete overlap in intraspecific, interspecific, and combined interspecific distances for *A. bipustulatus* in both the $p/q/p_{\text{lwr}}/p_{\text{upr}}$ and $p'/q'/p'_{\text{lwr}}/p'_{\text{upr}}$ directions since the metrics and their intervals attain magnitudes very close or equal to to one, with minimal noise (**Table 2**). As a result, this likely indicates that no DNA barcode gap is present for this species. Such findings are strongly reinforced by the very tight clustering of posterior draws (**Figure 5**) and associated interval estimates owing to the large number of specimens sampled for this species.

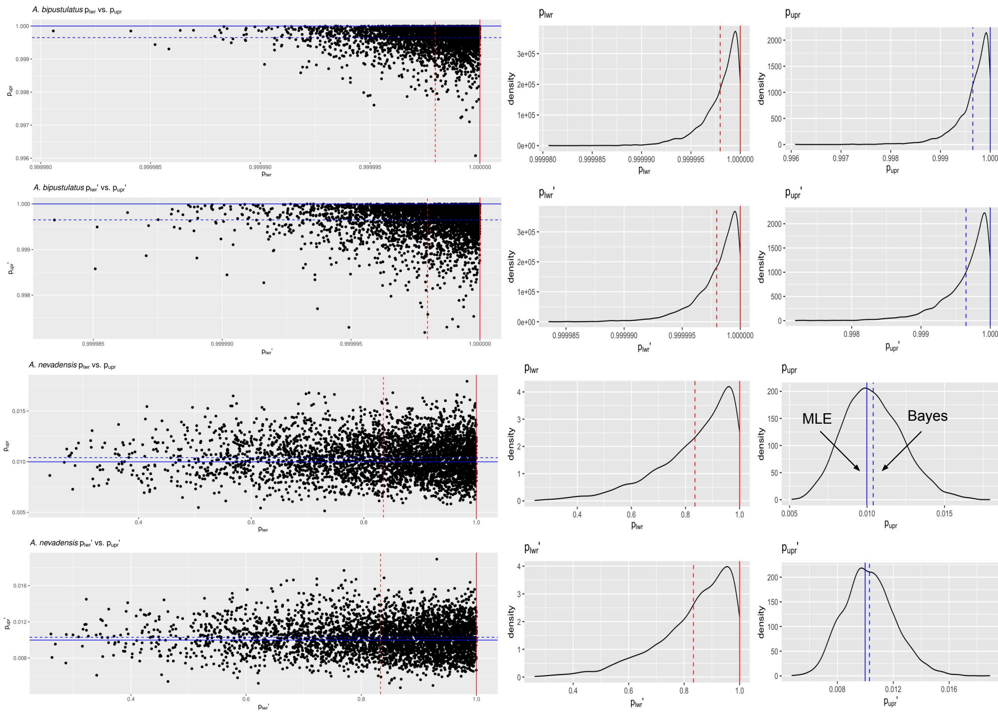

Figure 5: Scatterplots (black solid points) and distributions (black solid lines) depicting the DNA barcode gap metrics for *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) across CYTB based on 4000 Bayesian posterior draws. MLEs and posterior means are displayed as coloured (red/blue) solid and dashed lines for the metrics, respectively.

On the other hand, the situation for *A. nevadensis* is more nuanced, as posterior values are further spread out (**Table 2** and **Figure 5**), suggesting less overall certainty in true parameter values given the low specimen sampling coverage for this taxon. Of note, calculated SEs and SDs are large, with the 95% CIs and 95% CrIs being quite wide in most cases, consistent with much uncertainty regarding the computed frequentist and Bayesian posterior means of the DNA barcode gap metrics. For instance, the Bayesian analysis suggests that the data are consistent with both p_{lwr} and $p^{'}_{\text{lwr}}$ ranging from approximately 0.500-1.000 due to the large SD of about 0.140. The posterior means associated with these estimates themselves are also far from one. The CrI for p_{upr} and $p^{'}_{\text{upr}}$ also spans an order of magnitude, with the lower tail estimate of 0.007 being very close to zero. Further, regarding the frequentist analysis for the same species, the 95% CI for q_x is quite wide, reflecting considerable uncertainty in its true parameter value. Similarly, that for $q^{'}_x$ extends to negative values at the left endpoint, due to the corresponding SE of 0.070 being too high as a result of the extremely low sample size of $n = 2$ individuals sampled (**Table 2**). Since the 95% CI for $q_x^{'}$ truncated at the lower endpoint includes the value of zero, the null hypothesis for the presence of a DNA barcode gap would not be rejected. Despite this, it is worth noting that truncation is not standard statistical practice and will likely lead to an interval with less than 95% nominal coverage having high relative error. In such cases, more appropriate confidence interval methods like the Wilson score interval, the exact (Clopper-Pearson) interval, or the Agresti-Coull interval should be employed (Newcombe, 1998; Agresti and Coull, 1998). KDEs for *A. bipustulatus* are strongly left (negatively) skewed (**Figure 5**), whereas those for *A. nevadensis* exhibit more symmetry, especially for p_{upr} and $p_{\text{upr}}^{'}$ (**Figure 5**). These differences are likely due to the stark contrast in sample sizes for the two examined species. Nevertheless, simulated counts of overlapping specimen records from the posterior predictive distribution (**Table 3**) were found to be very close to observed counts for both species, indicating that the proposed model adequately captures underlying variation. Despite this, while CrIs were found to be narrow for *A. bipustulatus*, they were quite wide for *A. nevadensis* (particularly for y_{upr} and $y_{\text{upr}}^{'}$), suggestng that further model improvements could be made.

Table 3: Posterior predictive checks of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ($N = 701$) and *A. nevadensis* ($N = 2$) for CYTB. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

Species	Variable	*Y*	*n*	Bayes Est. (SD; 95% CrI)
A. bipustulatus	y_{lwr}	491401.000	491401.000	491400.018 (1.378; 491396.000-491401.000)
A. bipustulatus	y_{upr}	2804.000	2804.000	2803.019 (1.433; 2799.000-2804.000)
A. bipustulatus	$y_{\text{lwr}}^{'}$	491401.000	491401.000	491400.008 (1.412; 491396.000-491401.000)
A. bipustulatus	y_{upr}	2804.000	2804.000	2802.992 (1.429; 2799.000-2804.000)
A. nevadensis	y_{lwr}	4.000	4.000	3.355 (0.888; 1.000-4.000)
A. nevadensis	y_{upr}	28.000	2804.000	29.151 (7.620; 16.000-45.000)
A. nevadensis	$y_{\text{lwr}}^{'}$	4.000	4.000	3.325 (0.884; 1.000-4.000)
A. nevadensis	y_{upr}	28.000	2804.000	28.942 (7.409; 15.000-45.000)

Obtained results suggest that use of the Beta(1, 1) prior may not be appropriate given a low number of collected individuals for most taxa in DNA barcoding efforts. This suggests that further consideration of more informative beta priors is worthwhile. This is clear for *A.*

nevadensis, where use of a stronger prior will likely eliminate the issue regarding accurate estimation of p_{lwr} and p'_{lwr} and stabilize their uncertainty.

4 Conclusion

Herein, the accuracy of the DNA barcode gap was analyzed from a rigorous statistical lens to expedite both the curation and growth of reference sequence libraries, ensuring they are populated with high quality, statistically defensible specimen records fit for purpose to address standing questions in ecology, evolutionary biology, biodiversity forensics, conservation, management, and restoration. To accomplish this, recently proposed, easy to calculate, nonparametric MLEs were formally derived using ECDFs/CECDFs and applied to assess the extent of overlap/separation of distance distributions within and among two species of predatory water beetles in the genus *Agabus* sequenced at CYTB using a Bayesian binomial count model with conjugate beta priors. Findings highlight a high level of parameter uncertainty for *A. nevadensis*, whereas posterior estimates of the DNA barcode gap metrics for *A. bipustulatus* are much more certain. Based on these results, it is imperative that specimen sampling be prioritized to better reflect actual species boundaries. More generally, apart from the metrics being employed to better highlight the importance of within-species genetic diversity versus between-species divergence, it is expected that the approach developed herein will be of broad utility in applied fields, such as DNA-based detection of seafood fraud within global supply chains, and in the determination of species occupancy/detection probabilities at ecological sites of interest using active and passive environmental DNA (eDNA) methods such as metabarcoding.

Since the DNA barcode gap metrics often attain values very close to zero (suggesting no overlap and complete separation of distance distributions) and/or very near one (indicating no separation and complete overlap), in addition to more intermediate values, a noninformative Beta$(\frac{1}{2}, \frac{1}{2})$ prior may be more appropriate over complete ignorance imposed by a Beta(1, 1) prior. The former distribution is U-shaped symmetric and places greater probability density at the extremes of the distribution due to its heavier tails, while still allowing for variability in parameter estimates within intermediate values along its domain. This feature is appealing since Phillips et al.'s (2024) metrics often take on values of zero and one. Note that this prior is Jeffreys' prior density (Jeffreys, 1946), which is proportional to the square root of the Fisher information $\mathcal{I}(\theta)$; that is, $\pi(\theta) \propto \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}$. Jeffreys' prior has several desirable statistical properties as a prior: that it is inversely proportional to the standard deviation of the binomial distribution, and most notably, that it is invariant to model reparameterization (Gelman et al., 2014). However, this prior can lead to divergent transitions, among other pathologies, imposed by complex geometry (*i.e.*, curvature) in the posterior space since many iterative stochastic MCMC sampling algorithms experience difficulties when exploring high density distribution regions. Thus, remedies to resolve them, such as lowering the step size of the HMC/NUTS sampler, should be attempted in future work, along with other approaches such as empirical Bayes estimation to approximate beta prior hyperparameters from observed data through the MLE or other methods of parameter estimation, such as the method of moments. Alternatively, hierarchical modelling could be employed to estimate separate distribution model hyperparameters for each species and/or compute distinct estimates for the directionality/comparison level of the DNA barcode gap metrics (*i.e.,* lower *vs.* upper, non-prime *vs.* prime) separately within the genus under study. This would permit greater flexibility through incorporating more fine-grained structure seen in the data. However, low taxon sample sizes may preclude valid inferences to be reasonably ascertained due to the large number additional parameters which would be introduced through the specification of the hyperprior distributions. Methods outlined in Gelman et al. (2014, 2020), such as dealing with non-exchangeability of observations and alternate model parameterizations like the logit, may prove useful in this regard. Perhaps, the best course of action though is the elicitation of priors from domain experts. Even though more work remains, it is clear that both frequentist and Bayesian inference hold much promise for the future of molecular biodiversity science.

Supplementary Information

None declared

Data Availability Statement

Raw data, R, and Stan code can be accessed via Dryad at:
http://datadryad.org/stash/share/
RZIfMixcEODe0RWP7eyXWQewSVbqEIA9UTrH3ZVKyn4.

A GitHub repository can be found at:
https://github.com/jphill01/Bayesian-DNA-Barcode-Gap-Coalescent.

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Conflict of Interest

None declared.

Author Contributions

JDP wrote the manuscript, wrote R and Stan code, as well as analyzed and interpreted all model results. RHH and NH provided feedback on the final version of the manuscript.

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